Channel, sediment, nutrient

Purpose

These files report what happens in the channel network: flow, sediment and nutrients entering, leaving and stored in each channel, channel geometry and erosion, and sediment budgets for each channel, each stream order and the whole basin.

SWAT+ routes channels defined in the channel files of the SWAT+ Editor (see Channels and channel-lte). Two rows of `print.prt` control all of the channel files:

`print.prt` row	Files opened
`channel_sd`	`channel_sd_*`, `channel_sdmorph_*`, `sd_chanbud_*`, and the sediment budget files `ch_sedbud_*`, `chanbud_order_*` and `sedbud_*`. With a sub-daily time step and `daily` set to `y`, also `channel_sd_subday.txt`.
`basin_sd_cha`	`basin_sd_cha_*`, `basin_sd_chamorph_*`, `basin_sd_chanbud_*`

Each `y` in the `daily`, `monthly`, `yearly` or `avann` column of a row opens the matching file. Most files use the suffixes `_day`, `_mon`, `_yr` and `_aa`. The `ch_sedbud` and `chanbud_order` files use `_d`, `_m`, `_y` and `_a`. With `csvout` set to `y`, a `.csv` copy of each file is written as well.

The channel files are opened only when the project has at least one channel, except `sd_chanbud_*`, which is opened whenever its switch is on.

The `channel` and `basin_cha` rows open the legacy channel files described at the end of this page.

Files at a glance

File	One row per	Content
`channel_sd_*`	channel	Water surface, storage, inflow and outflow of water, sediment and nutrients, and stream temperature.
`channel_sdmorph_*`	channel	Flow rates, sediment transport and erosion, and channel width, depth and slope.
`sd_chanbud_*`	channel	Sediment and nutrient budget: inflow, outflow, deposition and erosion.
`ch_sedbud_*`	channel	Bank and bed erosion, floodplain deposition, channel size.
`chanbud_order_*`	stream order (1 to 12)	The `ch_sedbud` values summed over all channels of each stream order.
`sedbud_*`	basin (one row)	Basin sediment budget: upland yield, bank and bed erosion, floodplain, reservoir and wetland deposition.
`basin_sd_cha_*`	basin	`channel_sd` summed over all channels.
`basin_sd_chamorph_*`	basin	`channel_sdmorph` summed over all channels.
`basin_sd_chanbud_*`	basin	`sd_chanbud` summed over all channels.
`channel_sd_subday.txt`	channel and time step	Outflow for each sub-daily time step.

Leading columns

Files with one row per channel start with seven columns:

#	Column	Description
1	`jday`	Day of the year of the last day in the print step.
2	`mon`	Month.
3	`day`	Day of the month.
4	`yr`	Calendar year.
5	`unit`	Channel number.
6	`gis_id`	GIS identifier of the channel.
7	`name`	Channel name.

The basin files have the same seven columns with `unit` and `gis_id` set to 1 and the project name in `name`. In `ch_sedbud_*` the same three columns are labelled `ichan`, `ob` and `name`. `chanbud_order_*` has `iord` (the stream order) in place of all three, and `sedbud_*` has only the four date columns.

channel_sd files

After the leading columns come four values for the water surface, then three blocks of 18 values (storage, inflow, outflow) and a final stream temperature.

Column	Units	Description
`area`	ha	Water surface area of the channel.
`precip`	m^3	Precipitation falling on the water surface.
`evap`	m^3	Evaporation from the water surface.
`seep`	m^3	Seepage through the channel bed.

Each block has the same 18 quantities. The column names end in `_stor`, `_in` or `_out`:

Quantity	Storage units	Inflow and outflow units	Description
`flo`	m^3	m^3/s	Water: volume stored, or average flow rate.
`sed`	tons	tons	Sediment.
`orgn`	kg N	kg N	Organic nitrogen.
`sedp`	kg P	kg P	Organic (sediment-attached) phosphorus.
`no3`	kg N	kg N	Nitrate.
`solp`	kg P	kg P	Soluble mineral phosphorus.
`chla`	kg	kg	Chlorophyll a.
`nh3`	kg N	kg N	Ammonium.
`no2`	kg N	kg N	Nitrite.
`cbod`	kg	kg	Carbonaceous biochemical oxygen demand.
`dox`	kg	kg	Dissolved oxygen.
`san`	tons	tons	Sand.
`sil`	tons	tons	Silt.
`cla`	tons	tons	Clay.
`sag`	tons	tons	Small aggregates.
`lag`	tons	tons	Large aggregates.
`grv`	tons	tons	Gravel.
`null`	°C	°C	Water temperature.

The 18th column of each block is headed `null` and has no unit in the units row. It is the water temperature of the stored water, the inflow and the outflow. The outflow temperature is the stream temperature from the heat balance calculation.

The last column, `water_temp` (°C), is not the heat balance temperature. It is $5 + 0.75\,\bar{T}$, where $\bar{T}$ is the mean air temperature of the day at the weather station of the channel that was routed last. All channels therefore show the same value for a given day. Use the `null` column of the outflow block for the stream temperature of each channel.

Aggregation

- The storage block is the storage at the end of the print step, also in the monthly, yearly and average annual files.
- Inflow and outflow rates (`flo_in`, `flo_out`) are averaged over the days of the step. All other inflow and outflow quantities are totals for the step; the average annual file gives the average yearly total.
- Water temperatures in the inflow and outflow blocks are flow-weighted averages.
- `precip`, `evap` and `seep` are totals for the step. `area` is the average daily area in the monthly and yearly files.

channel_sdmorph files

Column	Units	Description
`flo_in`	m^3/s	Inflow rate.
`geo_bf`	m^3/s	Base flow from the geomorphic aquifer method.
`flo_out`	m^3/s	Outflow rate.
`peakr`	m^3/s	Peak flow rate.
`sed_in`	tons	Sediment entering the channel.
`sed_out`	tons	Sediment leaving the channel.
`washld`	tons	Wash load (suspended sediment) leaving the channel.
`bedld`	tons	Sediment deposited in the channel.
`dep`	tons	Sediment deposited on the floodplain.
`deg_btm`	tons	Bed erosion.
`deg_bank`	tons	Bank erosion.
`hc_sed`	tons	Sediment from gully headcut erosion.
`width`	m	Bankfull top width at the end of the step.
`depth`	m	Bankfull depth at the end of the step.
`slope`	m/m	Channel slope at the end of the step.
`deg_btm`	m	Bed downcutting.
`deg_bank`	m	Bank widening.
`hc_len`	m	Headcut retreat.
`flo_in_mm`	mm	Inflow as a depth over the channel's drainage area.
`aqu_in_mm`	mm	Geomorphic base flow as a depth over the drainage area.
`flo_out_mm`	mm	Outflow as a depth over the drainage area.
`sed_stor`	tons	Sediment stored in the channel.
`n_tot`	kg N	Total nitrogen leaving the channel (organic N, nitrate, ammonium and nitrite).
`p_tot`	kg P	Total phosphorus leaving the channel (organic and soluble P).
`dep_bf`	m	Flow depth at bankfull.
`velav_bf`	m/s	Average velocity at bankfull.

`deg_btm` and `deg_bank` each appear twice, first in tons and then in m; read them by position. The units row labels `n_tot` as kg_N, `p_tot` as kg_P and `dep_bf` as m/s; `dep_bf` is a depth in m.

In the monthly and yearly files, the flow rates, `sed_stor`, `n_tot`, `p_tot`, `dep_bf` and `velav_bf` are daily averages; the tons, mm and m erosion columns are totals; and `width`, `depth` and `slope` are the values at the end of the step. The average annual file gives daily averages for the first group and average yearly totals for the second.

sd_chanbud files

Column	Description
`in_sed`, `out_sed`	Sediment entering and leaving the channel.
`fp_dep`	Sediment deposited on the floodplain.
`ch_dep`	Sediment deposited in the channel.
`bank_ero`	Bank erosion.
`bed_ero`	Bed erosion.
`in_no3`, `in_orgn`, `out_no3`, `out_orgn`	Nitrate and organic nitrogen entering and leaving.
`fp_no3`	Nitrate lost on the floodplain.

Column	Description
bank_no3, bed_no3	Nitrate gained from bank and bed erosion.
fp_orgn, ch_orgn	Organic nitrogen deposited on the floodplain and in the channel.
bank_orgn, bed_orgn	Organic nitrogen gained from bank and bed erosion.
in_solp, in_orgp, out_solp, out_orgp	Soluble and organic phosphorus entering and leaving.
fp_solp	Soluble phosphorus lost on the floodplain.
bank_solp, bed_solp	Soluble phosphorus gained from bank and bed erosion.
fp_orgp, ch_orgp	Organic phosphorus deposited on the floodplain and in the channel.
bank_orgp, bed_orgp	Organic phosphorus gained from bank and bed erosion.
no3_orgn	Nitrate converted to organic nitrogen in the channel.
solp_orgp	Soluble phosphorus converted to organic phosphorus in the channel.

The sediment columns (in_sed to bed_ero) are in tons. The nitrogen and phosphorus columns are in kg, although the units row shows tons for every column. All values are totals for the print step; the average annual file gives the average yearly total.

Sediment budget files

These three file groups are new in 62.0.1. They use the same channel erosion and deposition values, summed in different ways. Their rows are written in free format (list-directed), so column widths vary.

ch_sedbud files (per channel)

Column	Units	Description
num	-	Number of days in the step on which the channel had inflow.
wid	m	Bankfull width, summed over those days.
dep	m	Bankfull depth, summed over those days.
fp_km2	km^2	Floodplain area, summed over those days. The floodplain is taken as five times the channel width over the channel length.
ebank_t	tons	Bank erosion.
ebtm_t	tons	Bed erosion.
ebank_m	m	Bank erosion as a width.
ebtm_m	m	Bed erosion as a depth.
fp_t	tons	Floodplain deposition.
fp_mm	mm	Floodplain deposition as a depth over the floodplain, assuming a bulk density of 1 t/m^3.

wid, dep and fp_km2 are summed over days, so divide them by num to get the average width, depth and floodplain area in the daily, monthly and yearly files. In the average annual file the sums are divided by the number of printed years but num is not, so the ratio is too small by that factor.

chanbud_order files (per stream order)

One row for each stream order from 1 to 12 in every step, with the ch_sedbud columns summed over all channels of that order. num is then the number of channel-days with inflow, and wid / num and dep / num give the average width and depth for the order. fp_mm is summed over channels and is not a depth for the order as a whole. The stream order of each channel comes from the order column of the channel database.

sedbud files (basin)

Column	Units	Description
upland_t	tons	Sediment yield from all HRUs.
ebank_t	tons	Bank erosion in all channels.
ebtm_t	tons	Bed erosion in all channels.
fp_t	tons	Floodplain deposition in all channels.

Column	Units	Description
res_dep_t	tons	Net deposition in all reservoirs (sediment in minus sediment out).
wet_dep_t	tons	Net deposition in all wetlands.

Deposition inside the channels themselves is not part of this file; it is in the ch_dep column of sd_chanbud.

Print timing

The three sediment budget file groups start after the nyskip warm-up years, like the other files. Their daily files have a row for every day after that and do not follow the daily print window (day_start to day_end, interval) in print.prt.

Basin files

basin_sd_cha_*, basin_sd_chamorph_* and basin_sd_chanbud_* have the same columns as channel_sd_*, channel_sdmorph_* and sd_chanbud_*, with one row for the basin. Each value is the sum over all channels in the basin. Flows are summed along the whole network, so flo_out in basin_sd_cha is not the flow leaving the basin; use the outlet channel's row in channel_sd for that. basin_sd_cha has no water_temp column, and in basin_sd_chamorph the width, depth and slope columns are those of the last channel.

Sub-daily outflow

When the simulation uses a sub-daily time step and the channel_sd row has y in daily, channel_sd_subday.txt has one row per channel per time step: jday, mon, day, yr, unit, gis_id, tstep (time step number), name and flo_out. flo_out is the volume of water (m^3) leaving the channel in that time step, although the header gives m^3/s. The units row and the header of the .csv copy are those of the channel_sdmorph file.

Legacy channel files

The channel and basin_cha rows of print.prt open channel_*.txt and basin_cha_*.txt for legacy channel objects (chan in object.cnt). Their 59 columns cover flow, sediment, nutrients, pesticides, bacteria, metals, sediment size classes and erosion:

```
flo_in flo_out evap tloss sed_in sed_out sed_conc orgn_in orgn_out orgp_in orgp_out
no3_in no3_out nh4_in nh4_out no2_in no2_out solp_in solp_out chla_in chla_out
cbod_in cbod_out dis_in dis_out solpst_in solpst_out sorbpst_in sorbpst_out react
volat setlpst resuspst difus reactb bury sedpest bacp baclp met1 met2 met3 sand_in
sand_out silt_in silt_out clay_in clay_out smag_in smag_out lag_in lag_out grvl_in
grvl_out bnk_ero ch_deg ch_dep fp_dep tot_ssed
```

SWAT+ 62.0.1 does not route legacy channels (the routing call is disabled), so these files contain only zeros. Projects built with the SWAT+ Editor use the channels described above.

Example

The columns of every chanbud_order_* file:

```
jday  mon  day  yr  iord  num  wid  dep  fp_km2  ebank_t  ebtm_t  ebank_m  ebtm_m  fp_t  fp_mm
```

Take 40 first-order channels, each 5 m wide, that have flow on every day of a 10-year print period. In chanbud_order_y.txt the row with iord 1 has num = 40 × 365 = 14,600 and wid = 73,000, so wid / num gives the 5 m average width. In chanbud_order_a.txt the same row has num = 146,000 (the count over all ten years) and wid = 73,000 (the ten-year sum divided by 10), so the average width is 10 × wid / num.

The reference outputs in refdata/Ames_sub1 of the SWAT+ repository were written by revision 61.0.1 and contain none of the sediment budget files.

Related files

- print.prt: switches each file on or off.
- Channels and channel-lte: channel inputs, including the stream `order` column.
- Water balance outputs: the landscape water that reaches the channels.
- Aquifer, reservoir, plant outputs: reservoir and wetland files.
- Output naming convention: all output files.

Sources

- File opening and headers: header_sd_channel.f90, header_write.f90, header_channel.f90
- Header and record types: sd_channel_module.f90, hydrograph_module.f90, water_body_module.f90, channel_module.f90, `basin_sediment_budget` in basin_module.f90
- Channel writers: sd_channel_output.f90, sd_chanmorph_output.f90, sd_chanbud_output.f90
- Sediment budget writers: ch_sedbud_output.f90, chord_sedbud_output.f90, basin_sedbud_output.f90
- Basin writers: basin_sdchannel_output.f90, basin_chanmorph_output.f90, basin_chanbud_output.f90, basin_channel_output.f90
- Values: sd_channel_control3.f90, sd_channel_sediment3.f90, ch_temp.f90; upland, reservoir and wetland terms in hru_control.f90, res_control.f90, wetland_control.f90
- Call order: command.f90

File	Header type	Record
`channel_sd_*`	`ch_wbod_hdr`, `hyd_stor_hdr`, `hyd_in_hdr`, `hyd_out_hdr`, `wtmp_hdr`	`ch_wat_*`, `ch_stor`, `ch_in_*`, `ch_out_*`, `wtemp`
`channel_sdmorph_*`	`sdch_hdr`	`chsd_*` (`sd_ch_output`)
`sd_chanbud_*`	`sdch_bud_hdr`	`ch_sed_bud*` (`channel_sediment_budget_output`)
`ch_sedbud_*`	`ch_sedbud_hdr`	`ch_morph*` (`channel_morphology_output`)
`chanbud_order_*`	`ch_bud_order_hdr`	`ch_morph_ord*`
`sedbud_*`	`ch_sed_bud_hdr`	`bsn_sedbud*` (`basin_sediment_budget`)
`channel_sd_subday.txt`	`sdch_hdr_subday`	`ob%hyd_flo(1,ii)`

Column	Fortran variable
`geo_bf`	`sd_ch_output%aqu_in`
`hc_len`	`sd_ch_output%hc_m`
`flo_out_mm`	`sd_ch_output%flo_mm`
second `deg_btm`, `deg_bank`	`deg_btm_m`, `deg_bank_m`
`null` (each block)	`hyd_output%temp`
`water_temp`	module variable `wtemp`
`ob` (`ch_sedbud`)	`ob(iob)%gis_id`